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## **A Project Report on Simulation of BGP Prefix Hijacking, Traffic Engineering, and Policy-Based Mitigation in a Multi- ISP Mini-Internet Using GNS3**

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# Abstract

This project builds and evaluates a six–Autonomous System (AS) “mini-Internet” in GNS3 using Cisco c7200 routers to simulate realistic BGP behavior. The topology includes a Tier-1 transit provider, two ISPs, a multi-homed legitimate prefix owner (victim), a customer network, and an attacker AS. We first established normal eBGP connectivity and verified global reachability between customer and victim networks. We then simulated two BGP prefix hijacking attacks: (1) a basic origin hijack of a /24 and (2) a more-specific /25 hijack that exploits longestprefix-match forwarding behavior. Finally, we implemented layered policy-based mitigations including prefix filtering, AS-PATH filtering, and max-prefix protection, and demonstrated traffic engineering using Local Preference and AS-PATH prepending to reflect business routing decisions.

Tested results show that without filtering, the attacker’s announcements can become the selected BGP best path in parts of the network, and the /25 hijack consistently diverts traffic for addresses within the more specific block. After applying customer/provider policies (prefixlists + route-maps, AS-PATH constraints, and max-prefix limits), hijacked routes were rejected and end-to-end traffic from the customer LAN to the victim LAN was restored. Traffic engineering changes successfully shifted preferred outbound and inbound paths according to policy, demonstrating how routing decisions can reflect economics and operational goals in addition to shortest-path metrics.

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## Chapter 1

### Introduction

The Border Gateway Protocol (BGP) is the backbone routing protocol of the global Internet, responsible for exchanging reachability information between Autonomous Systems (ASes). Despite its critical role, BGP was designed under a trust-based model and lacks built-in mechanisms to verify the authenticity of route advertisements. This architectural limitation has made BGP vulnerable to misconfigurations and malicious attacks, most notably prefix hijacking.

In recent years, numerous real-world incidents have demonstrated how incorrect or malicious BGP announcements can divert traffic, disrupt services, or enable traffic interception. These incidents highlight the importance of understanding BGP behavior and implementing operational safeguards at ISP and enterprise boundaries.

This project focuses on simulating a realistic multi-ISP environment using GNS3 to study normal BGP operation, analyze the impact of prefix hijacking attacks, and evaluate the effectiveness of commonly deployed mitigation techniques. By emulating real Cisco IOS routers, the project aims to bridge theoretical routing concepts with practical, observable behavior.

## 1.1 Goal of the Project

The goal of this project is to simulate, observe, and mitigate Border Gateway Protocol (BGP) prefix hijacking within a realistic multi-ISP mini-Internet environment. The project demonstrates normal eBGP route propagation across multiple interconnected Autonomous Systems and examines how routing behavior changes under adversarial conditions. Two common hijacking variants are analyzed: a same-length prefix hijack involving a /24 network and a morespecific /25 hijack that exploits longest-prefix-match forwarding behavior. In addition, the project evaluates defense-in-depth mitigation techniques deployed at ISP boundaries, including prefix filtering, AS-PATH filtering, and maximum prefix limits. Traffic engineering mechanisms using Local Preference and AS-PATH prepending are also implemented to illustrate how routing decisions are influenced by operational and business policies rather than purely technical metrics.

## 1.2 Summary of Work and Findings

The project begins by establishing a stable multi-AS BGP environment and verifying legitimate end-to-end connectivity between customer and victim networks. Controlled hijacking scenarios are then introduced from a malicious AS, and their impact on routing tables, path selection, and traffic forwarding is observed. Experimental results show that a basic /24 origin hijack may succeed partially depending on BGP best-path selection and network topology. In contrast, a more-specific /25 hijack consistently attracts traffic for the covered address range due to longest-prefix-match forwarding. The results further demonstrate that strict prefix filtering at provider edges is the most effective practical mitigation. AS-PATH filtering and max-prefix limits provide complementary protection and contribute to a layered security posture. Finally, traffic engineering experiments confirm that Local Preference can override AS-PATH length, allowing operators to implement routing decisions based on policy and business relationships.

# Chapter 2

## Objectives of the Project

The primary objective of this project is to design and implement a multi-Autonomous System mini-Internet and evaluate BGP routing behavior under both normal and adversarial conditions. The project aims to demonstrate how legitimate BGP routing is established across multiple ISPs and how insufficient routing policies can allow unauthorized or malicious route advertisements to propagate through the network. Additional objectives include simulating realistic BGP prefix hijacking attacks, analyzing their effects on routing tables, best-path selection, and reachability, and implementing layered mitigation techniques such as prefix filtering, AS-PATH filtering, and maximum prefix limits. The project also seeks to demonstrate policy-driven routing decisions through traffic engineering mechanisms that reflect real-world operational practices.



# Chapter 3

## Background and Motivation

### 3.1 Overview of BGP

The Border Gateway Protocol is a path-vector interdomain routing protocol responsible for exchanging reachability information between Autonomous Systems. Unlike interior routing protocols, BGP does not compute routes based on shortest-path metrics. Instead, it applies a structured decision process that prioritizes policy attributes such as Local Preference, AS-PATH length, and origin type. This design allows network operators to enforce routing policies aligned with business agreements, but it also introduces complexity and security challenges.

### 3.2 BGP Prefix Hijacking

BGP prefix hijacking occurs when an Autonomous System advertises an IP prefix that it does not legitimately own. If other networks accept the advertisement, traffic destined for the legitimate prefix owner may be diverted, intercepted, or dropped. While many hijacking incidents result from accidental misconfigurations, deliberate attacks have also been observed. Morespecific prefix hijacks, such as advertising a /25 instead of a legitimate /24, are particularly effective because IP forwarding follows the longest-prefix-match rule, causing traffic to prefer the attacker's route even when the legitimate route remains available.

### 3.3 Why GNS3 (Emulation) for This Work

GNS3 was selected as the experimental platform because it supports emulation of real Cisco IOS routing behavior, including eBGP session handling, policy enforcement, and convergence dynamics. Unlike abstract simulators, GNS3 allows direct interaction with IOS configuration features such as prefix-lists, route-maps, and AS-PATH filters, making it well suited for studying operational BGP security mechanisms in a controlled environment.

# Chapter 4

## Literature Review

Border Gateway Protocol (BGP) security has been an active area of research and operational concern since the early expansion of the global Internet. Because BGP was designed without

intrinsic authentication or validation mechanisms, numerous studies and operational reports have documented its vulnerability to route leaks, misconfigurations, and deliberate prefix hijacking. This literature review summarizes relevant concepts and operational practices directly related to BGP hijacking and mitigation, which form the foundation of the experimental work in this project.

## 4.1 BGP Routing and Trust Model

BGP operates on a trust-based interdomain routing model in which Autonomous Systems exchange reachability information with limited inherent verification. Early studies highlight that BGP assumes cooperative behavior among network operators, relying on bilateral agreements and manual policy enforcement to maintain routing correctness. While this model enabled rapid Internet scalability, it also introduced systemic weaknesses, as routers generally accept routing announcements from neighbors unless explicitly filtered.

Research and operational analyses emphasize that BGP decision-making prioritizes policy attributes such as Local Preference and AS-PATH length over shortest-path metrics. As a result, routing behavior is highly dependent on operator-defined policies, making misconfigurations or malicious announcements capable of influencing large portions of the routing system.

## 4.2 BGP Prefix Hijacking

Prefix hijacking is a well-documented routing security incident in which an Autonomous System announces an IP prefix that it does not legitimately own. Studies distinguish between origin hijacks, where the attacker claims ownership of an existing prefix, and more-specific prefix hijacks, where the attacker announces a longer prefix to exploit longest-prefix-match forwarding. Operational incident analyses consistently show that more-specific hijacks are particularly effective, as forwarding behavior prefers the most specific route regardless of BGP best-path selection for less specific prefixes.

Real-world cases reported by network operators and regional Internet registries demonstrate that hijacks can lead to traffic interception, service disruption, or large-scale outages. These findings directly motivate the attack scenarios implemented in this project.

## 4.3 Operational Mitigation Techniques

Operational best practices for mitigating BGP hijacking focus on policy enforcement at network boundaries. Prefix filtering, based on customer or peer authorization, is widely regarded as the most effective first line of defense, as it prevents unauthorized announcements from entering the routing system. AS-PATH filtering provides additional validation by restricting acceptable origin AS values for known prefixes, though it depends on accurate knowledge of routing relationships.

Max-prefix limits are commonly deployed to protect routers from excessive route advertisements caused by configuration errors or malicious behavior. While max-prefix

mechanisms do not prevent hijacking directly, they reduce operational impact by limiting routing table growth and session instability.

More recent literature introduces cryptographic solutions such as Resource Public Key Infrastructure (RPKI) and Route Origin Validation (ROV), which aim to formally verify prefix ownership. Although these mechanisms represent a significant advancement in routing security, their deployment remains partial. Consequently, traditional policy-based mitigations remain highly relevant and are the focus of this project's experimental evaluation.

## Chapter 5

### Problem Statement

Despite widespread deployment of BGP, many networks operate with minimal filtering and validation of routing information. This project investigates how easily BGP prefix hijacking can occur in a multi-ISP environment and how effectively standard operational practices mitigate it.

#### 5.1 Description

Design and implement a multi-AS network that:

- Demonstrates normal BGP operations across interconnected ISPs and customers.
- Simulates two hijack attacks (same-prefix and more-specific).
- Implements layered mitigation techniques.
- Demonstrates traffic engineering using policy attributes.

#### 5.2 Calculations

##### Subnetting for /30 Point-to-Point Links

A /30 provides 4 addresses: network, two usable, broadcast. Example: 10.0.46.0/30

- Network: 10.0.46.0
- Hosts: 10.0.46.1, 10.0.46.2
- Broadcast: 10.0.46.3

### /24 and /25 Prefix Behavior

- 203.0.113.0/24 provides 256 addresses (254 usable).
- 203.0.113.0/25 covers 203.0.113.0–203.0.113.127 (128 addresses).
- Forwarding uses longest-prefix-match; thus /25 is preferred over /24 for covered destinations.

## 5.3 Software and Hardware

Software: GNS3 (Windows), Dynamips c7200 IOS (advipservices), VPCS.

Hardware: 8 GB RAM recommended, stable CPU; Idle-PC required for c7200 performance.

# Chapter 6

## Designs

### 6.1 AS Roles and Prefix Ownership

Table 6.1: Autonomous System Roles

| AS Number | Role                  | Notes                                     |
|-----------|-----------------------|-------------------------------------------|
| AS100     | Customer network      | Originates 198.51.100.0/24                |
| AS200     | Attacker AS           | Hijacks 203.0.113.0/24 and 203.0.113.0/25 |
| AS300     | Regional ISP          | Aggregation + policy enforcement point    |
| AS400     | Victim / Legit owner  | Originates 203.0.113.0/24 (multi-homed)   |
| AS500     | Tier-1 transit        | Transit between regions                   |
| AS600     | Upstream ISP of AS400 | Provider path for AS400                   |

Table 6.2: Prefix Ownership

| AS    | Prefix                                                      |
|-------|-------------------------------------------------------------|
| AS400 | 203.0.113.0/24 (victim/legitimate)                          |
| AS100 | 198.51.100.0/24 (customer)                                  |
| AS200 | Unauthorized advertisements: 203.0.113.0/24, 203.0.113.0/25 |

### 6.2 IP Addressing Plan

Table 6.3: Inter-AS Links (Point-to-Point)

| Link      | Subnet       | Purpose         |
|-----------|--------------|-----------------|
| R400–R600 | 10.0.46.0/30 | AS400–<br>AS600 |
| R600–R500 | 10.0.56.0/30 | AS600–<br>AS500 |
| R500–R300 | 10.0.53.0/30 | AS500–<br>AS300 |
| R300–R100 | 10.0.31.0/30 | AS300–<br>AS100 |

|           |              |                 |
|-----------|--------------|-----------------|
| R300–R200 | 10.0.32.0/30 | AS300–<br>AS200 |
| R300–R400 | 10.0.34.0/30 | AS300–<br>AS400 |

Table 6.4: LAN Networks

| LAN       | Prefix          | Gateway                  |
|-----------|-----------------|--------------------------|
| AS100 LAN | 198.51.100.0/24 | 198.51.100.1 (R100 f0/0) |
| AS400 LAN | 203.0.113.0/24  | 203.0.113.1 (R400 f0/0)  |

## 6.3 BGP Design

All inter-AS BGP sessions are eBGP. AS400 is multi-homed to AS300 and AS600. The attacker AS200 peers to AS300.

## 6.4 Policy/Mitigation Design

- Prefix filtering: Accept only authorized prefixes from neighbors (especially customers).
- AS-PATH filtering: Ensure expected origin AS for critical prefixes (lab policy).
- Max-prefix: Limit number of accepted prefixes to reduce blast radius of leaks/floods.

# Chapter 7

## Network Design and Topology

### 7.1 Overall Architecture

The simulated network represents a small-scale but realistic interdomain Internet environment composed of six Autonomous Systems interconnected through eBGP sessions.

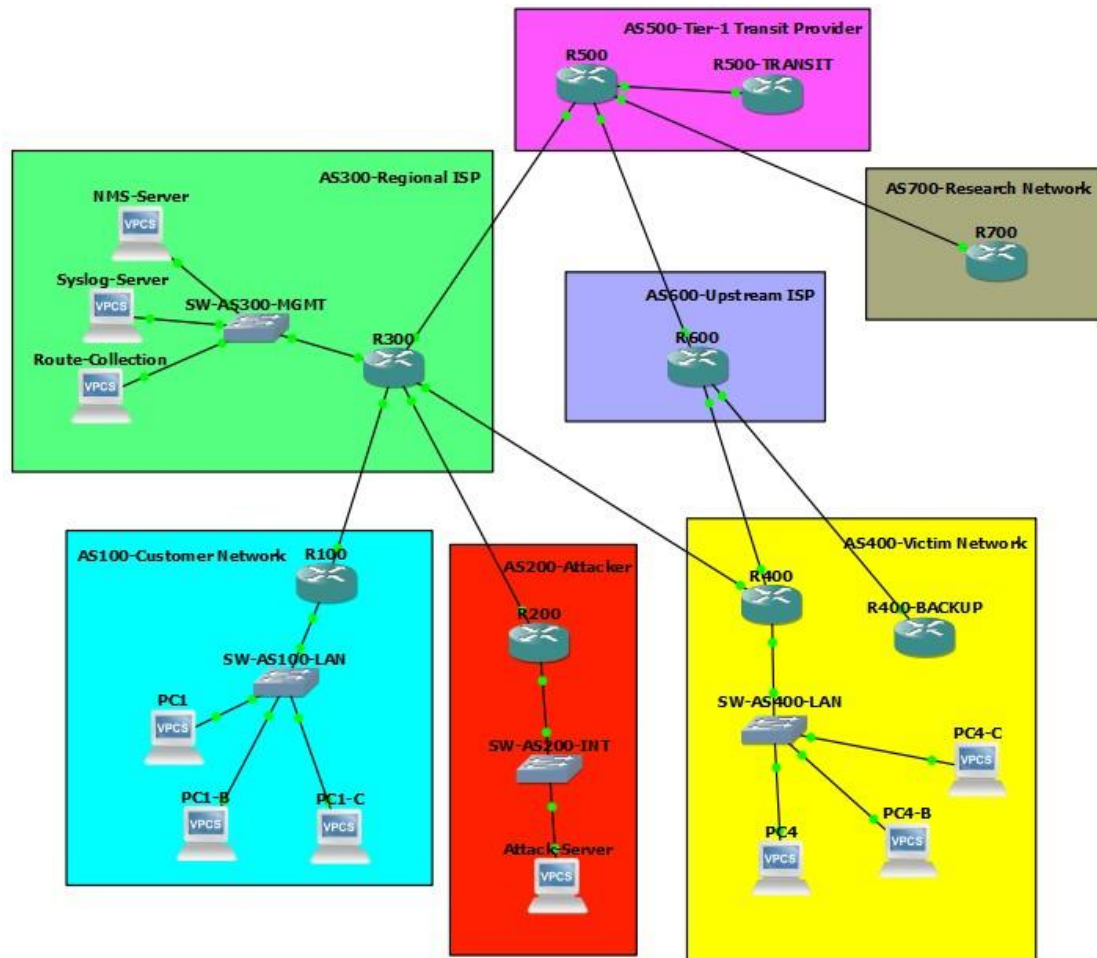


Figure 7.1: Multi-AS Mini-Internet Topology in GNS3

## 7.2 Roles of Autonomous Systems

Table 7.1: Autonomous System Roles and Functions

| AS Number | Role             | Description                        |
|-----------|------------------|------------------------------------|
| AS100     | Customer Network | Originates customer prefix         |
| AS200     | Attacker         | Announces illegitimate prefixes    |
| AS300     | Regional ISP     | Aggregation and policy enforcement |
| AS400     | Victim           | Legitimate prefix owner            |
| AS500     | Tier-1 Provider  | Internet transit                   |
| AS600     | Upstream ISP     | Primary provider for AS400         |

## 7.3 IP Addressing Scheme

Table 7.2: Summary of IP Address Allocation

| Link / Network | IP Subnet       |
|----------------|-----------------|
| AS400 – AS600  | 10.0.46.0/30    |
| AS600 – AS500  | 10.0.56.0/30    |
| AS500 – AS300  | 10.0.53.0/30    |
| AS300 – AS100  | 10.0.31.0/30    |
| AS300 – AS200  | 10.0.32.0/30    |
| AS300 – AS400  | 10.0.34.0/30    |
| AS100 LAN      | 198.51.100.0/24 |
| AS400 LAN      | 203.0.113.0/24  |

## Chapter 8

# Implementation

### 8.1 Simulation Environment

The network was implemented using GNS3 with Cisco IOS c7200 routers running the advipservices image and VPCS hosts. An appropriate Idle-PC value was applied to each c7200 router to prevent excessive CPU utilization and ensure stable simulation performance. All routers were configured with IP routing enabled and basic interface configurations before establishing BGP sessions.

### 8.2 Normal BGP Operation (Verification)

Key commands used:

```
show ip bgp summary
show ip bgp show ip bgp 203.0.113.0 ping
203.0.113.10 traceroute
203.0.113.10
```



```

R300#show ip bgp
BGP table version is 4, local router ID is 10.0.53.2
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
               r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
               x best-external, a additional-path, c RIB-compressed,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

   Network          Next Hop              Metric LocPrf Weight Path
*> 198.51.100.0      10.0.31.2                0           0 100 i
*> 203.0.113.0       10.0.34.2                0           0 400 i
*                   10.0.53.1                0 500 600 400 i
R300#show ip bgp 203.0.113.0
BGP routing table entry for 203.0.113.0/24, version 3
Paths: (2 available, best #1, table default)
  Advertised to update-groups:
    1
  Refresh Epoch 2
  400
    10.0.34.2 from 10.0.34.2 (203.0.113.1)
      Origin IGP, metric 0, localpref 100, valid, external, best
      rx pathid: 0, tx pathid: 0x0
  Refresh Epoch 2
  500 600 400
    10.0.53.1 from 10.0.53.1 (10.0.56.2)
      Origin IGP, localpref 100, valid, external
      rx pathid: 0, tx pathid: 0
R300#

```

Figure 8.1: show ip bgp summary on AS300 During Normal Operation

```

R100#show ip bgp
BGP table version is 3, local router ID is 198.51.100.1
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
               r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
               x best-external, a additional-path, c RIB-compressed,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

   Network          Next Hop              Metric LocPrf Weight Path
*> 198.51.100.0      0.0.0.0                  0          32768 i
*> 203.0.113.0       10.0.31.1                0           0 300 400 i
R100#show ip route bgp
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       + - replicated route, % - next hop override

Gateway of last resort is not set

B    203.0.113.0/24 [20/0] via 10.0.31.1, 00:12:12
R100#

```

Figure 8.2: show ip bgp summary on AS100 During Normal Operation

```

R500#show ip bgp
BGP table version is 3, local router ID is 10.0.56.2
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
               r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
               x best-external, a additional-path, c RIB-compressed,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

   Network          Next Hop              Metric LocPrf Weight Path
* > 198.51.100.0     10.0.53.2                      0 300 100 i
*   203.0.113.0      10.0.53.2                      0 300 400 i
* > 10.0.56.1        10.0.56.1                      0 600 400 i
R500#

```

Figure 8.3: show ip bgp summary on AS500 During Normal Operation

## 8.3 Attack Scenarios

### 8.3.1 Attack 1: Basic Prefix Hijack (/24)

The attacker (AS200) originates the victim's prefix 203.0.113.0/24 using BGP. A static route to Null0 is added so IOS can advertise the prefix via the network statement.

```

R300#show ip bgp 203.0.113.0
BGP routing table entry for 203.0.113.0/24, version 3
Paths: (3 available, best #2, table default)
  Advertised to update-groups:
    1
  Refresh Epoch 1
  200
    10.0.32.2 from 10.0.32.2 (10.0.32.2)
      Origin IGP, metric 0, localpref 100, valid, external
      rx pathid: 0, tx pathid: 0
  Refresh Epoch 2
  400
    10.0.34.2 from 10.0.34.2 (203.0.113.1)
      Origin IGP, metric 0, localpref 100, valid, external, best
      rx pathid: 0, tx pathid: 0x0
  Refresh Epoch 2
  500 600 400
    10.0.53.1 from 10.0.53.1 (10.0.56.2)
      Origin IGP, localpref 100, valid, external
      rx pathid: 0, tx pathid: 0
R300#

```

Figure 8.4: Evidence of Basic Prefix Hijack in BGP Table in AS300

```

R100#show ip bgp 203.0.113.0
BGP routing table entry for 203.0.113.0/24, version 3
Paths: (1 available, best #1, table default)
  Not advertised to any peer
  Refresh Epoch 2
  300 400
    10.0.31.1 from 10.0.31.1 (10.0.53.2)
      Origin IGP, localpref 100, valid, external, best
      rx pathid: 0, tx pathid: 0x0
R100#show ip route 203.0.113.0
Routing entry for 203.0.113.0/24
  Known via "bgp 100", distance 20, metric 0
  Tag 300, type external
  Last update from 10.0.31.1 00:26:29 ago
  Routing Descriptor Blocks:
    * 10.0.31.1, from 10.0.31.1, 00:26:29 ago
      Route metric is 0, traffic share count is 1
      AS Hops 2
      Route tag 300
      MPLS label: none
R100#

```

Figure 8.5: Evidence of Basic Prefix Hijack in BGP Table in AS100

```

R200#show ip bgp
BGP table version is 5, local router ID is 10.0.32.2
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
               r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
               x best-external, a additional-path, c RIB-compressed,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

   Network        Next Hop           Metric LocPrf Weight Path
* > 198.51.100.0   10.0.32.1           0         300 100 i
* > 203.0.113.0    0.0.0.0             0        32768 i
*                  10.0.32.1           0         300 400 i
R200#

```

Figure 8.6: Evidence of Basic Prefix Hijack in BGP Table in AS200

```

PC1> ping 203.0.113.10
84 bytes from 203.0.113.10 icmp_seq=1 ttl=61 time=90.559 ms
84 bytes from 203.0.113.10 icmp_seq=2 ttl=61 time=91.281 ms
84 bytes from 203.0.113.10 icmp_seq=3 ttl=61 time=111.314 ms
84 bytes from 203.0.113.10 icmp_seq=4 ttl=61 time=103.447 ms
84 bytes from 203.0.113.10 icmp_seq=5 ttl=61 time=83.068 ms

```

Figure 8.7: Evidence of Basic Prefix Hijack in BGP Table in PC1



### 8.3.2 Attack 2: More-Specific Prefix Hijack (/25)

The attacker announces 203.0.113.0/25. Due to longest-prefix-match, traffic destined to addresses within 203.0.113.0–203.0.113.127 is forwarded using the /25 route if present, making this attack highly effective.

```
R300#show ip bgp
BGP table version is 5, local router ID is 10.0.53.2
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
               r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
               x best-external, a additional-path, c RIB-compressed,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

   Network          Next Hop              Metric LocPrf Weight Path
*> 198.51.100.0      10.0.31.2                0         0 100 i
*> 203.0.113.0/25    10.0.32.2                0         0 200 i
* 203.0.113.0        10.0.32.2                0         0 200 i
*>                   10.0.34.2                0         0 400 i
*                    10.0.53.1                0 500 600 400 i
R300#
```

Figure 8.8: More-Specific Hijack and Forwarding Impact-AS300

```
R400#show ip bgp
BGP table version is 4, local router ID is 203.0.113.1
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
               r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
               x best-external, a additional-path, c RIB-compressed,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

   Network          Next Hop              Metric LocPrf Weight Path
* 198.51.100.0      10.0.46.1                0 600 500 300 100 i
*>                   10.0.34.1                0 300 100 i
* 203.0.113.0/25    10.0.46.1                0 600 500 300 200 i
*>                   10.0.34.1                0 300 200 i
*> 203.0.113.0       0.0.0.0                  0         32768 i
R400#
```

Figure 8.9: More-Specific Hijack and Forwarding Impact-AS400

```
R200#show ip bgp
BGP table version is 6, local router ID is 10.0.32.2
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
               r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
               x best-external, a additional-path, c RIB-compressed,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

   Network          Next Hop              Metric LocPrf Weight Path
*> 198.51.100.0      10.0.32.1                0 300 100 i
*> 203.0.113.0/25    0.0.0.0                  0         32768 i
*> 203.0.113.0       0.0.0.0                  0         32768 i
*                    10.0.32.1                0 300 400 i
R200#
```

Figure 8.10: More-Specific Hijack and Forwarding Impact-AS200

```

R100#show ip route 203.0.113.10
Routing entry for 203.0.113.0/25
  Known via "bgp 100", distance 20, metric 0
  Tag 300, type external
  Last update from 10.0.31.1 00:02:32 ago
  Routing Descriptor Blocks:
    * 10.0.31.1, from 10.0.31.1, 00:02:32 ago
      Route metric is 0, traffic share count is 1
      AS Hops 2
      Route tag 300
      MPLS label: none

R100#show ip route 203.0.113.200
Routing entry for 203.0.113.0/24
  Known via "bgp 100", distance 20, metric 0
  Tag 300, type external
  Last update from 10.0.31.1 00:37:41 ago
  Routing Descriptor Blocks:
    * 10.0.31.1, from 10.0.31.1, 00:37:41 ago
      Route metric is 0, traffic share count is 1
      AS Hops 2
      Route tag 300
      MPLS label: none
R100#

```

Figure 8.11: More-Specific Hijack and Forwarding Impact Observed at AS100

```

PC1> ping 203.0.113.10
*10.0.32.2 icmp_seq=1 ttl=253 time=62.097 ms (ICMP type:3, code:1, Destination host unreachable)
*10.0.32.2 icmp_seq=2 ttl=253 time=77.154 ms (ICMP type:3, code:1, Destination host unreachable)
*10.0.32.2 icmp_seq=3 ttl=253 time=76.237 ms (ICMP type:3, code:1, Destination host unreachable)
*10.0.32.2 icmp_seq=4 ttl=253 time=75.961 ms (ICMP type:3, code:1, Destination host unreachable)
*10.0.32.2 icmp_seq=5 ttl=253 time=75.741 ms (ICMP type:3, code:1, Destination host unreachable)
PC1>

```

Figure 8.12: More-Specific Hijack and Forwarding Impact-PC1

## Chapter 9

# Experimental and Theoretical Results

## 9.1 Sequence Chart (Control-Plane Events)

Normal sequence (simplified):

1. eBGP sessions established over TCP.
2. AS100 announces 198.51.100.0/24 to AS300.
3. AS400 announces 203.0.113.0/24 to AS300 and AS600.
4. Routes propagate via AS500 and across the topology.
5. PC1 reaches PC4 with successful ping/traceroute.

## 9.2 Outcome Summary Table

Table 9.1: Routing Behavior Under Different Scenarios

| Scenario                     | Selected/Installed<br>(Typical)             | Route | Outcome                                      |
|------------------------------|---------------------------------------------|-------|----------------------------------------------|
| Normal Operation             | Legit route from AS400<br>present globally  |       | Legitimate routing;<br>PC1→PC4 works         |
| Basic /24 Hijack             | AS200 route may become<br>best in some ASes |       | Partial diversion/blackholing<br>possible    |
| More-Specific<br>Hijack      | /25 AS200 /25 installed<br>covered IPs      | for   | Strong diversion for 203.0.113.0–<br>127     |
| After Filtering              | Only authorized routes<br>accepted          |       | Correct routing restored;<br>hijacks blocked |
| After Traffic<br>Engineering | Policy-driven preference<br>enforced        | Path  | preferences align with<br>business goals     |

## 9.3 Discussion

### 9.3.1 Why /25 Hijack Works Reliably

Forwarding uses longest-prefix-match. Even if the victim's /24 remains installed, the /25 is chosen for destinations it covers. This is a key reason more-specific hijacks are widely used.

### 9.3.2 Why Filtering is Effective

Provider edges act as trust boundaries. If AS300 and AS600 only accept authorized prefixes, malicious announcements are rejected early and cannot propagate widely.

# Chapter 10

## Mitigation Techniques

### 10.1 Layer 1: Prefix Filtering

Prefix filtering permits only expected prefixes from a neighbor. This is the strongest operational control in many ISP environments.

Table 10.1: Prefix Filtering Policy Summary

| Device | Neighbor          | Policy                      |
|--------|-------------------|-----------------------------|
| R300   | AS400 (10.0.34.2) | Accept only 203.0.113.0/24  |
| R300   | AS100 (10.0.31.2) | Accept only 198.51.100.0/24 |
| R300   | AS200 (10.0.32.2) | Deny all (lab attacker)     |
| R600   | AS400 (10.0.46.2) | Accept only 203.0.113.0/24  |

```
R300#show ip bgp
BGP table version is 6, local router ID is 10.0.53.2
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
               r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
               x best-external, a additional-path, c RIB-compressed,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

   Network        Next Hop        Metric LocPrf Weight Path
  *> 198.51.100.0   10.0.31.2         0         0 100 i
  *> 203.0.113.0    10.0.34.2         0         0 400 i
  *                10.0.53.1         0         0 500 600 400 i

R300#show ip prefix-list
ip prefix-list AS100_ONLY: 1 entries
  seq 10 permit 198.51.100.0/24
ip prefix-list AS400_ONLY: 1 entries
  seq 10 permit 203.0.113.0/24
R300#
```

Figure 10.1: Prefix Filtering on Routers



```
PC1> ping 203.0.113.10
84 bytes from 203.0.113.10 icmp_seq=1 ttl=61 time=91.293 ms
84 bytes from 203.0.113.10 icmp_seq=2 ttl=61 time=77.776 ms
84 bytes from 203.0.113.10 icmp_seq=3 ttl=61 time=91.753 ms
84 bytes from 203.0.113.10 icmp_seq=4 ttl=61 time=91.194 ms
84 bytes from 203.0.113.10 icmp_seq=5 ttl=61 time=76.109 ms
```

Figure 10.2: Output After Prefix Filtering on PC1

## 10.2 Layer 2: AS-PATH Filtering

AS-PATH filtering enforces expected origin ASes (lab policy). Example: accept 203.0.113.0/24 only if it originates in AS400.

## 10.3 Layer 3: Max-Prefix Protection

Max-prefix prevents route floods and limits session abuse by capping how many prefixes a neighbor can send.

```
R300#show running-config | section router bgp
router bgp 300
  bgp log-neighbor-changes
  neighbor 10.0.31.2 remote-as 100
  neighbor 10.0.31.2 description eBGP to R200 AS200
  neighbor 10.0.31.2 route-map FILTER_AS100_STRICT in
  neighbor 10.0.31.2 maximum-prefix 5 80
  neighbor 10.0.32.2 remote-as 200
  neighbor 10.0.32.2 route-map FILTER_AS200 in
  neighbor 10.0.34.2 remote-as 400
  neighbor 10.0.34.2 description eBGP to R400 AS400
  neighbor 10.0.34.2 route-map FILTER_AS400_STRICT in
  neighbor 10.0.34.2 maximum-prefix 5 80
  neighbor 10.0.53.1 remote-as 500
  neighbor 10.0.53.1 description eBGP to R500 AS500
R300#
```

Figure 10.3: Routers After Mitigation Techniques

```
R300#show ip prefix-list
ip prefix-list AS100_ONLY: 1 entries
    seq 10 permit 198.51.100.0/24
ip prefix-list AS400_ONLY: 1 entries
    seq 10 permit 203.0.113.0/24
R300#
```

```
R300#show ip as-path-access-list
AS path access list 10
    permit ^400$
AS path access list 20
    permit ^100$
R300#
```

```
R300#show ip bgp summary
BGP router identifier 10.0.53.2, local AS number 300
BGP table version is 6, main routing table version 6
2 network entries using 288 bytes of memory
3 path entries using 240 bytes of memory
3/2 BGP path/bestpath attribute entries using 408 bytes of memory
3 BGP AS-PATH entries using 72 bytes of memory
0 BGP route-map cache entries using 0 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
BGP using 1008 total bytes of memory
BGP activity 3/1 prefixes, 5/2 paths, scan interval 60 secs
```

| Neighbor  | V | AS  | MsgRcvd | MsgSent | TblVer | InQ | OutQ | Up/Down  | State/PfxRcd |
|-----------|---|-----|---------|---------|--------|-----|------|----------|--------------|
| 10.0.31.2 | 4 | 100 | 93      | 95      | 6      | 0   | 0    | 01:18:08 | 1            |
| 10.0.32.2 | 4 | 200 | 95      | 93      | 6      | 0   | 0    | 01:16:13 | 0            |
| 10.0.34.2 | 4 | 400 | 109     | 102     | 6      | 0   | 0    | 01:24:32 | 1            |
| 10.0.53.1 | 4 | 500 | 109     | 103     | 6      | 0   | 0    | 01:28:26 | 1            |

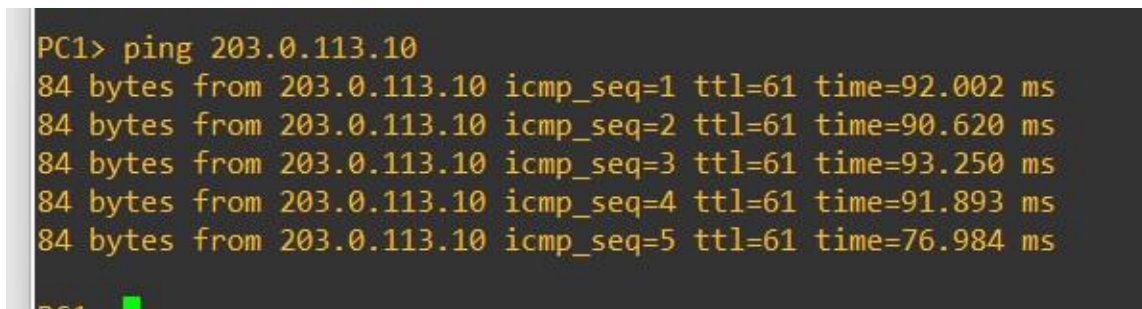
R300#

```
R300#show ip bgp
BGP table version is 6, local router ID is 10.0.53.2
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
               r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
               x best-external, a additional-path, c RIB-compressed,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found
```

| Network         | Next Hop  | Metric | LocPrf | Weight | Path          |
|-----------------|-----------|--------|--------|--------|---------------|
| *> 198.51.100.0 | 10.0.31.2 | 0      |        | 0      | 100 i         |
| *> 203.0.113.0  | 10.0.34.2 | 0      |        | 0      | 400 i         |
| *               | 10.0.53.1 |        |        | 0      | 500 600 400 i |

R300#

Figure 10.4: Routers After Mitigation Techniques



A terminal window with a dark background and yellow text. The prompt is 'PC1>'. The command entered is 'ping 203.0.113.10'. The output shows five successful ping responses, each 84 bytes from 203.0.113.10 with a TTL of 61 and varying times. A green cursor is visible on the line following the last ping result.

```
PC1> ping 203.0.113.10
84 bytes from 203.0.113.10 icmp_seq=1 ttl=61 time=92.002 ms
84 bytes from 203.0.113.10 icmp_seq=2 ttl=61 time=90.620 ms
84 bytes from 203.0.113.10 icmp_seq=3 ttl=61 time=93.250 ms
84 bytes from 203.0.113.10 icmp_seq=4 ttl=61 time=91.893 ms
84 bytes from 203.0.113.10 icmp_seq=5 ttl=61 time=76.984 ms
```

Figure 10.5: PC1 After Mitigation Techniques



# Chapter 11

## Traffic Engineering

### 11.1 Local Preference (Outbound Control)

Local Preference is used to influence outbound traffic selection. In this project, AS400 prefers AS600 as primary by setting higher local preference on routes learned from AS600.

```
R400#show ip bgp 198.51.100.0
BGP routing table entry for 198.51.100.0/24, version 8
Paths: (2 available, best #1, table default)
  Advertised to update-groups:
    1          2
  Refresh Epoch 2
  600 500 300 100
    10.0.46.1 from 10.0.46.1 (10.0.56.1)
      Origin IGP, localpref 200, valid, external, best
      rx pathid: 0, tx pathid: 0x0
  Refresh Epoch 3
  300 100
    10.0.34.1 from 10.0.34.1 (10.0.53.2)
      Origin IGP, localpref 50, valid, external
      rx pathid: 0, tx pathid: 0
R400#
```

```
R400#show ip route bgp
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       + - replicated route, % - next hop override

Gateway of last resort is not set

B      198.51.100.0/24 [20/0] via 10.0.46.1, 00:15:27
R400#
```

Figure 11.1: Local Preference

### 11.2 AS-PATH Prepending (Inbound Influence)

AS400 prepends its AS multiple times when advertising to AS300 to make the AS300 path less attractive for inbound selection.

```
R300#show ip bgp 203.0.113.0
BGP routing table entry for 203.0.113.0/24, version 7
Paths: (1 available, best #1, table default)
  Advertised to update-groups:
    1
  Refresh Epoch 3
  500 600 400
  10.0.53.1 from 10.0.53.1 (10.0.56.2)
    Origin IGP, localpref 100, valid, external, best
    rx pathid: 0, tx pathid: 0x0
R300#
```

Figure 11.2: AS-PATH Prepending

## 11.3 BGP Decision Process (Short Reference)

Common decision order:

1. Weight (higher wins)
2. Local Preference (higher wins)
3. Locally originated routes
4. AS-PATH length (shorter wins)
5. Origin type (IGP > EGP > Incomplete)
6. MED (lower wins)
7. eBGP over iBGP
8. Lowest IGP metric to next hop
9. Oldest path
10. Lowest router ID / lowest neighbor IP

## Chapter 12

# Conclusion

This project demonstrated the vulnerability of BGP to prefix hijacking and the effectiveness of policy-based mitigations in a realistic multi-AS mini-Internet. A basic /24 hijack can partially succeed depending on best-path selection, while a more-specific /25 hijack is highly effective due to longest-prefix-match forwarding. Implementing defense-in-depth (prefix filtering, AS-PATH filtering, max-prefix) blocked malicious routes and restored correct routing. Traffic engineering using Local Preference and AS-PATH prepending illustrated how routing decisions can reflect business policy rather than shortest paths.

# Chapter 13

## Future Work

Future extensions:

- Implement RPKI/ROV conceptually and compare with policy filtering.
- Add route leak prevention policies (customer-to-provider export rules).
- Add monitoring/alerting (syslog-based detection of unexpected origin changes).
- Expand topology to include more ASes and simulate route leaks vs hijacks.
- Add BGP communities for advanced policy tagging and blackholing.

# Bibliography

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- [3] Cisco Systems, *BGP Configuration Guide*, Cisco IOS Documentation. (Topics: prefix-lists, route-maps, AS-PATH access-lists, maximum-prefix limits)
- [4] MANRS, *Mutually Agreed Norms for Routing Security*, Internet Society. Available: <https://www.manrs.org>
- [5] RIPE NCC, *Routing Security and BGP Hijacking Case Studies*, RIPE Network Coordination Centre.



# Appendix A

## Addressing and Interface Assignment

Table A.1: Interfaces and IP Assignments

| Device | Interface | IP/Mask          | Peer/Use             |
|--------|-----------|------------------|----------------------|
| R400   | e1/0      | 10.0.46.2/30     | to R600 e1/0         |
| R400   | e1/1      | 10.0.34.2/30     | to R300 e1/3         |
| R400   | f0/0      | 203.0.113.1/24   | Victim LAN gateway   |
| R600   | e1/0      | 10.0.46.1/30     | to R400 e1/0         |
| R600   | e1/1      | 10.0.56.1/30     | to R500 e1/0         |
| R500   | e1/0      | 10.0.56.2/30     | to R600 e1/1         |
| R500   | e1/1      | 10.0.53.1/30     | to R300 e1/0         |
| R300   | e1/0      | 10.0.53.2/30     | to R500 e1/1         |
| R300   | e1/1      | 10.0.31.1/30     | to R100 e1/0         |
| R300   | e1/2      | 10.0.32.1/30     | to R200 e1/0         |
| R300   | e1/3      | 10.0.34.1/30     | to R400 e1/1         |
| R100   | e1/0      | 10.0.31.2/30     | to R300 e1/1         |
| R100   | f0/0      | 198.51.100.1/24  | Customer LAN gateway |
| R200   | e1/0      | 10.0.32.2/30     | to R300 e1/2         |
| PC1    | e0        | 198.51.100.10/24 | GW 198.51.100.1      |
| PC4    | e0        | 203.0.113.10/24  | GW 203.0.113.1       |

# Appendix B

## Verification Commands

### Interface and IP Verification

```
show ip interface brief show ip  
route connected ping <neighbor-
```

## ip> BGP Verification

```
show ip bgp summary show ip bgp  
show ip bgp 203.0.113.0 show ip bgp  
203.0.113.0/25 Policy
```

## Verification

```
show ip prefix-list show route-map show ip as-path-access-list  
show running-config | section router bgp show ip bgp neighbors
```

```
<neighbor-ip> routes Traffic Tests
```

```
ping 203.0.113.10 traceroute  
203.0.113.10 ping 198.51.100.10  
traceroute 198.51.100.10
```

# Appendix C

## Full Router and VPCS Configurations (Divided by Phases)

### Important Note on IOS BGP **network** Statements

For IOS, a network X mask Y statement requires that the exact prefix exists in the routing table. Legitimate LANs are connected routes. Hijacked prefixes on AS200 are created using static routes to Null0 so they can be originated.

### Phase 3: Base Configuration (All Routers)

```
enable configure terminal  
no ip domain-lookup  
! hostname RXXX (set per router)  
end write  
memory
```

## Phase 2: IP Addressing

### R100

enable configure terminal

hostname R100 no ip

domain-lookup

interface e1/0 description Link to R300 ip address  
10.0.31.2 255.255.255.252 no shutdown

interface f0/0 description AS100 LAN ip address  
198.51.100.1 255.255.255.0 no shutdown

end write

memory

### R200

enable configure terminal

hostname R200 no ip

domain-lookup

interface e1/0 description Link to R300 ip address  
10.0.32.2 255.255.255.252 no shutdown

end write

memory

### R300

enable configure terminal

hostname R300 no ip

domain-lookup

interface e1/0 description Link to R500 ip address  
10.0.53.2 255.255.255.252 no shutdown

interface e1/1 description Link to R100 ip address  
10.0.31.1 255.255.255.252 no shutdown

```
interface e1/2 description Link to R200 ip address
 10.0.32.1 255.255.255.252 no shutdown
```

```
interface e1/3 description Link to R400 ip address
 10.0.34.1 255.255.255.252 no shutdown
```

```
end write
memory
```

## R400

```
enable configure terminal
hostname R400 no ip
domain-lookup
```

```
interface e1/0 description Link to R600 ip address
 10.0.46.2 255.255.255.252 no shutdown
```

```
interface e1/1 description Link to R300 ip address
 10.0.34.2 255.255.255.252 no shutdown
```

```
interface f0/0 description AS400 LAN ip address
 203.0.113.1 255.255.255.0 no shutdown
```

```
end write
memory
```

## R500

```
enable configure terminal
hostname R500 no ip
domain-lookup
```

```
interface e1/0 description Link to R600 ip address
 10.0.56.2 255.255.255.252 no shutdown
```

```
interface e1/1 description Link to R300 ip address
 10.0.53.1 255.255.255.252 no shutdown
```

```
end write
memory
```

## R600

```
enable configure terminal
hostname R600 no ip
domain-lookup
```

```
interface e1/0 description Link to R400 ip address
10.0.46.1 255.255.255.252 no shutdown
```

```
interface e1/1 description Link to R500 ip address
10.0.56.1 255.255.255.252 no shutdown
```

```
end write
memory
```

## VPCS

```
PC1 ip 198.51.100.10 255.255.255.0 198.51.100.1 save
```

```
PC4 ip 203.0.113.10 255.255.255.0 203.0.113.1 save
```

## Phase 4: Legitimate BGP (Normal Operation)

### R100 (AS100)

```
enable configure terminal router bgp 100 bgp log-neighbor-
changes network 198.51.100.0 mask 255.255.255.0
neighbor 10.0.31.1 remote-as 300 end write memory
```

### R200 (AS200) – No advertisements yet

```
enable configure terminal router bgp 200 bgp
log-neighbor-changes neighbor 10.0.32.1
remote-as 300 end write memory
```

### R300 (AS300)

```
enable configure terminal router bgp 300 bgp
log-neighbor-changes neighbor 10.0.53.1
remote-as 500 neighbor 10.0.31.2 remote-as 100
neighbor 10.0.32.2 remote-as 200 neighbor
10.0.34.2 remote-as 400 end write memory
```

## R400 (AS400) – Victim / Legit Owner

```
enable configure terminal router bgp 400 bgp log-neighbor-changes network 203.0.113.0 mask 255.255.255.0 neighbor 10.0.46.1 remote-as 600 neighbor 10.0.34.1 remote-as 300 end write memory
```

## R500 (AS500)

```
enable configure terminal router bgp 500 bgp log-neighbor-changes neighbor 10.0.56.1 remote-as 600 neighbor 10.0.53.2 remote-as 300 end write memory
```

## R600 (AS600)

```
enable configure terminal router bgp 600 bgp log-neighbor-changes neighbor 10.0.46.2 remote-as 400 neighbor 10.0.56.2 remote-as 500 end write memory
```

## Phase 5: Attack Scenario 1 (Basic /24 Hijack) – On R200

```
enable configure terminal ip route 203.0.113.0 255.255.255.0 Null0 router bgp 200 network 203.0.113.0 mask 255.255.255.0 end write memory
```

## Phase 6: Attack Scenario 2 (More-Specific /25 Hijack) – On R200

```
enable configure terminal ip route 203.0.113.0 255.255.255.128 Null0 router bgp 200 network 203.0.113.0 mask 255.255.255.128 end write memory
```

## Phase 7: Mitigation Layer 1 (Prefix Filtering)

### On R300

```
enable configure terminal
```

```
ip prefix-list AS100_ONLY seq 10 permit 198.51.100.0/24 ip prefix-list AS400_ONLY seq 10 permit 203.0.113.0/24
```

```
route-map FROM_AS100 permit 10 match ip address  
  prefix-list AS100_ONLY
```

```
route-map FROM_AS400 permit 10 match ip address  
  prefix-list AS400_ONLY
```

```
route-map FROM_AS200 deny 10
```

```
router bgp 300 neighbor 10.0.31.2 route-map FROM_AS100 in  
  neighbor 10.0.34.2 route-map FROM_AS400 in neighbor  
  10.0.32.2 route-map FROM_AS200 in
```

```
end write  
memory
```

```
clear ip bgp 10.0.31.2 soft in clear ip bgp 10.0.34.2  
soft in clear ip bgp 10.0.32.2 soft in
```

## On R600 (extra protection from AS400)

```
enable configure terminal
```

```
ip prefix-list AS400_LEGIT seq 10 permit 203.0.113.0/24 route-map  
FROM_AS400_UP permit 10 match ip address prefix-list AS400_LEGIT
```

```
router bgp 600 neighbor 10.0.46.2 route-map FROM_AS400_UP in
```

```
end write  
memory
```

```
clear ip bgp 10.0.46.2 soft in
```

## Phase 8: Mitigation Layer 2 (AS-PATH Filtering) – On R300

```
enable configure terminal
```

```
ip as-path access-list 10 permit ^100$ ip as-path access-list  
20 permit ^400$
```

```
route-map FROM_AS100_STRICT permit 10
```

```

match ip address prefix-list AS100_ONLY match as-path 10

route-map FROM_AS400_STRICT permit 10 match ip
  address prefix-list AS400_ONLY match as-path 20

router bgp 300 neighbor 10.0.31.2 route-map FROM_AS100_STRICT in
  neighbor 10.0.34.2 route-map FROM_AS400_STRICT in

end write
memory

clear ip bgp * soft in

```

## Phase 9: Mitigation Layer 3 (Max-Prefix Protection) – On R300

```

enable configure terminal router bgp 300 neighbor 10.0.32.2 maximum-
prefix 1 warning-only neighbor 10.0.31.2 maximum-prefix 5 80
neighbor 10.0.34.2 maximum-prefix 5 80 end write memory

```

## Phase 10: Traffic Engineering

### Local Preference (Outbound Preference) – On R400

```

enable configure terminal

route-map PREFER_AS600 permit 10 set local-
  preference 200 route-map BACKUP_AS300
  permit 10 set local-preference 50

router bgp 400 neighbor 10.0.46.1 route-map PREFER_AS600 in
  neighbor 10.0.34.1 route-map BACKUP_AS300 in

end write
memory

clear ip bgp * soft in

```

### AS-PATH Prepending (Inbound Influence) – On R400

```

enable configure terminal

```



```
route-map PREPEND_TO_AS300 permit 10 set as-  
path prepend 400 400 400
```

```
router bgp 400 neighbor 10.0.34.1 route-map PREPEND_TO_AS300 out
```

```
end write  
memory
```

```
clear ip bgp 10.0.34.1 soft out
```

## GitHub Link

GitHub link: [https://github.com/Shafiullah-Raihan/CCNA\\_Project\\_BGP\\_Hijack](https://github.com/Shafiullah-Raihan/CCNA_Project_BGP_Hijack)